Question (1) (20 points) A. (5 points) Definitions

a. Define AI as Acting rationally

An agent is something that perceives and acts

• A rational agent is one that acts so as to achieve the best outcome

b. Define the state space search.

Graph of states (Express constraints and parameters of the problem)

c. What is the difference between omniscient and rational agents?

Omniscient	rational agents
An omniscient agent knows the actual outcome of its actions and can act accordingly; • omniscience is impossible in reality The omniscience of an agent knows the actual outcome assuming infinite knowledge	Rationality is distinct from omniscience (allknowing with infinite knowledge) -A rational agent is one that does the right thing

d- What does it mean for an inference rule to be sound? Does soundness imply completeness?

The conclusion is true whenever the premises are true. Soundness means that you cannot prove anything that's wrong. Completeness means that you can prove anything that's right.

e- What does it mean for a sentence to be satisfiable? Does Satisfiability imply validity?

A sentence is satisfiable if it is true under some interpretation (i.e. it has a model), otherwise the sentence is unsatisfiable.

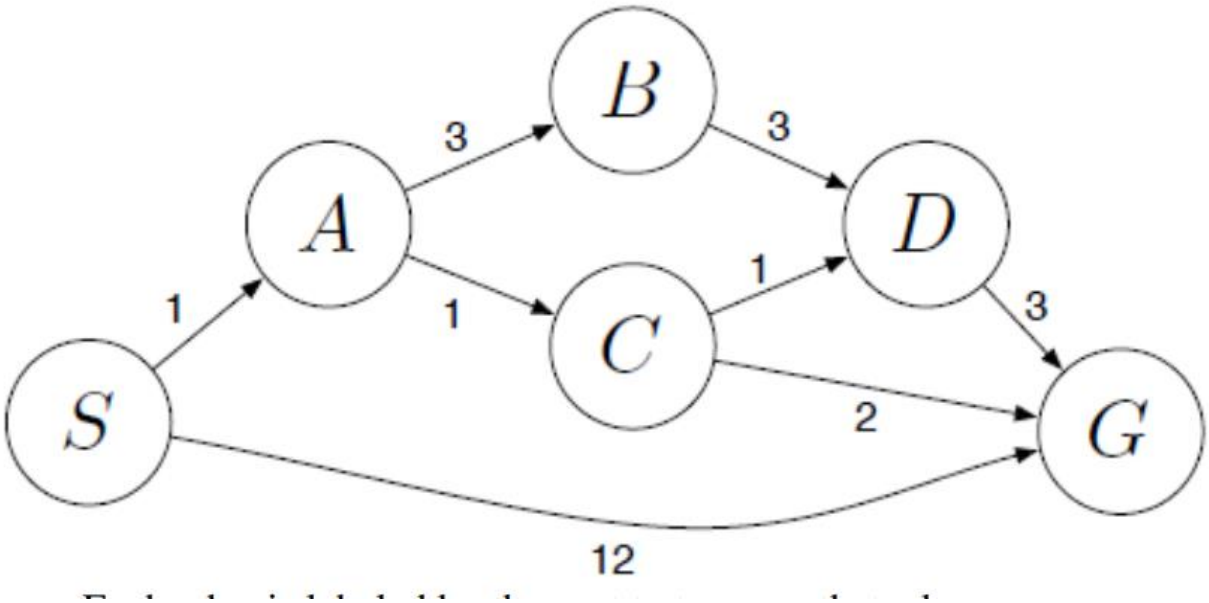
A sentence is valid if and only if its negation is unsatisfiable.

B. (8 points) Develop a PEAS description of the automated taxi-driver.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe: fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

Figure 2.4 PEAS description of the task environment for an automated taxi.

C. (7 points) Consider the following search space where S is the start state and G is the goal state:



Breadth First Algorithm:

Visited states in order: S, A, G

Path to the goal: S -> G

Depth First Algorithm:

Visited states in order: S, A, B, D, G

Path to the goal: S -> G

A Algorithm:*

Visited states in order: S, A, C, G

Path to the goal: S -> A -> C -> G

d- Is the h used above admissible or not? Why?

The heuristic function "h" used in the A* algorithm above is not admissible.

A: The heuristic estimate h(A) is 2, but the true optimal cost h*(A) is 7 (A to G).

C: The heuristic estim ate h(C) is 6, but the true optimal cost h*(C) is 5 (C to D to G).

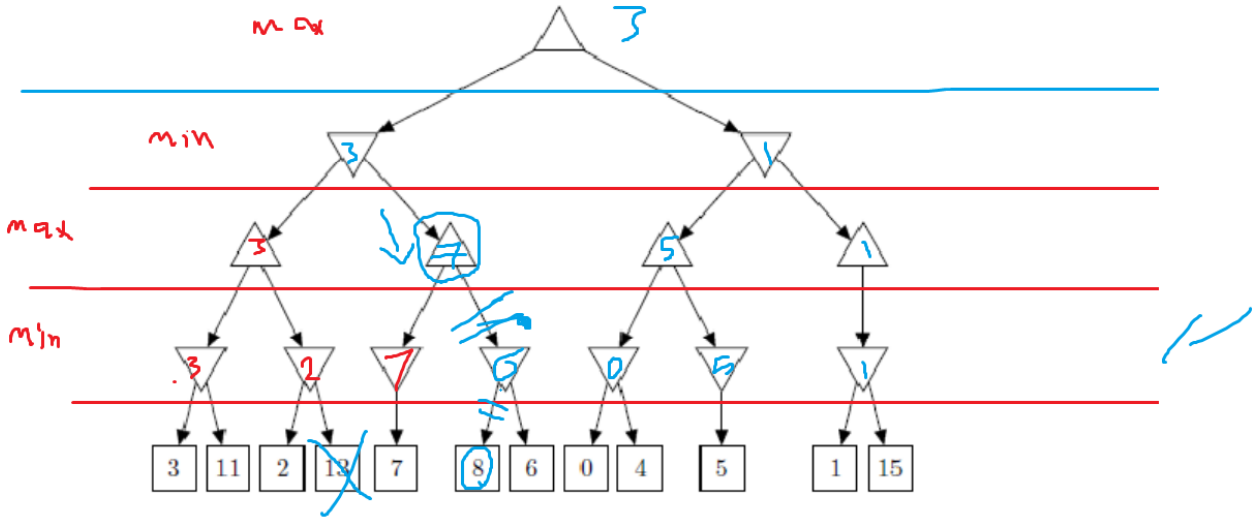
Question (2) (15 points)

A. (5 points) On the minimax game tree below, assuming its max's turn to play. Cross out the

B.

Question (2) (15 points)

A. **(5 points)** On the minimax game tree below, assuming its max's turn to play. Cross out the branches removed by alpha-beta pruning assuming left to right traversal.



branches removed by alpha-beta pruning assuming left to right traversal.

B. (4 points) Both Local beam search, genetic algorithms combine an uphill tendency with random exploration and exchange of information among parallel search threads.

a. Explain how this is done in both.

b. What advantage the genetic algorithm has over the local beam search?

Local beam search One Solution to improve hill Climbing.

• Keep track of k states instead of one –

Initially: k randomly selected states –

Next: determine all successors of k states –

If any of successors is goal $\rightarrow$ finished – Else select k best from successors and repeat.

GA emulates genetic evolution.

ν A GA has distinct features:

A string representation of chromosomes.

A selection procedure for initial population and for off-spring creation.

A cross-over method and a mutation method.

A fitness function.

A replacement procedure.

b- genetic algorithms have advantages over local beam search in terms of their ability to explore a broader search space, maintain a population of solutions, strike a balance between exploration and exploitation, support parallelism, and adapt to various problem types. These factors can make genetic algorithms more effective in finding optimal solutions, especially in complex, multi-modal search spaces.

C. (6 points) Consider the following fitness function: fitness = 5a + 3bc – d + 2e where a-e are all Boolean-valued parameters. Consider the following initial population

a	*b*	*c*	*d*	*e*
1	1	0	1	1
0	1	1	0	1
1	1	0	0	0
1	1	1	1	0
1	0	0	0	0

a. Compute the fitness of each of the members of the initial population above.

Member 1: (a=1, b=1, c=0, d=1, e=1)
fitness = 51 + 310 - 1 + 21 = 5 - 1 + 2 = 6
Member 2: (a=0, b=1, c=1, d=0, e=1)
fitness = 50 + 311 - 0 + 21 = 0 + 3 - 0 + 2 = 5
Member 3: (a=1, b=1, c=0, d=0, e=0)
fitness = 51 + 310 - 0 + 20 = 5 + 0 - 0 + 0 = 5
Member 4: (a=1, b=1, c=1, d=1, e=0)
fitness = 51 + 311 - 1 + 20 = 5 + 3 - 1 + 0 = 7
Member 5: (a=1, b=0, c=0, d=0, e=0)
fitness = 51 + 300 - 0 + 20 = 5 + 0 - 0 + 0 = 5

b-Assuming the best two (with greater fitness values) of members of the population are selected for reproduction, and the cross-over point is that between the b and the c, show the resulting children:

	one pointCross-over
Parent 1 : Parent 2 :	1 1 0 1 1 1 1 1 1 0
Child 1: Child 2:	1 1 1 1 0 1 1 0 1 1

Question (3) (25 points)

A. (3 points) What does it mean that FOL is semidecidable, how does that affect the resolution inference rule.

FOL (First-Order Logic) is semidecidable, meaning it has an algorithm to determine truth, but it may not always stop. This affects resolution theorem proving, which is sound and complete but may not always terminate due to FOL's non-termination possibility.

B. (5 points) Put in the CNF Form

i. $A \leftrightarrow (B \vee C)$

Step 1: Start with the biconditional formula A <--> (B v C).

Step 2: Apply the definition of the biconditional:
A <--> (B v C) is equivalent to (A --> (B v C)) ∧ ((B v C) --> A).

Step 3: Apply the implication rule (A --> B is equivalent to ¬A ∨ B):
(¬A ∨ (B v C)) ∧ (¬(B v C) ∨ A).

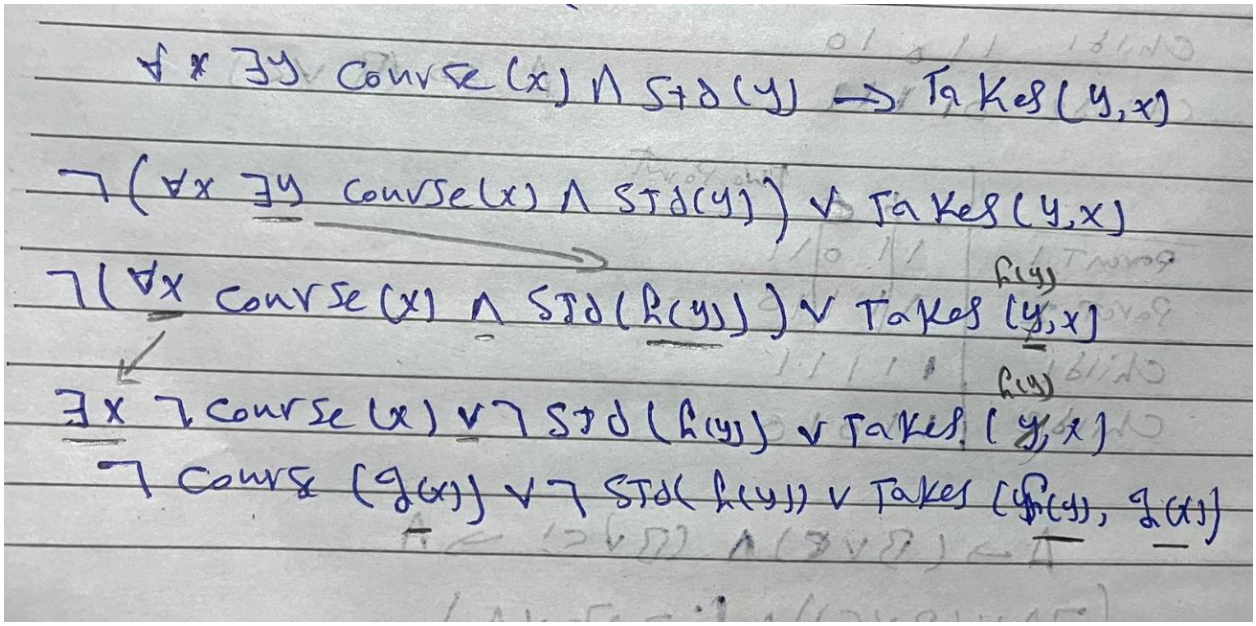
Step 4: توزيه النفي علي القوس:
(¬A ∨ B ∨ C) ∧ ((¬B ∧ ¬C) ∨ A).

Step 5: Apply De Morgan's law (¬(A ∧ B) is equivalent to (¬A ∨ ¬B)):
(¬A ∨ B ∨ C) ∧ ((¬B ∨ ¬C) ∨ A).

Step 6: Apply the commutative property of ∧ and ∨ to rearrange the terms:
(¬A ∨ B ∨ C) ∧ (A ∨ ¬B ∨ ¬C).

Step 7: The formula is now in CNF, where each clause is a disjunction of literals, and the entire formula is a conjunction of clauses:
(¬A ∨ B ∨ C) ∧ (A ∨ ¬B ∨ ¬C).

ii. $\forall x \exists y\ course(x) \wedge std(y) \rightarrow takes(y,x)$



C. (12 points) Represent the following sentences in FOL

i. All watches contain at least two buttons.

$\forall x\ (W(x) \rightarrow (\exists y1\ \exists y2\ (B(y1, x) \wedge B(y2, x) \wedge (y1 \neq y2))))$

This representation asserts that for any watch (x), there exist at least two distinct buttons (y1 and y2) on that watch.

ii. Only doctors use watches

$\forall x\ (D(x) \rightarrow W(x))$

This representation asserts that for all individuals (x), if x is a doctor (D(x)), then x uses a watch (W(x)).

iii. Ali has a blue watch

$\exists x\ (B(x) \wedge H(a, x))$

This representation asserts that there exists an individual (x) such that x is blue (B(x)) and Ali (a) has (H) that individual as a watch.

iv. There is exactly one pink watch.

$\exists x (P(x) \land \forall y (P(y) \rightarrow x = y))$

This representation asserts that there exists an individual (x) such that x is pink (P(x)), and for all individuals (y) that are pink, it implies that y is equal to x. This ensures that there is exactly one pink watch represented by x.

v. All watches must show 12 hours.

$\forall x (W(x) \rightarrow S(x, 12))$

This representation asserts that for all watches (x), if x is a watch (W(x)), then x shows 12 hours (S(x, 12)). It means that all watches must display 12 hours.

vi. Apple and Samsung watches have the same price.

$\forall x \forall y (\text{AppleWatch}(x) \land \text{SamsungWatch}(y) \rightarrow \text{SamePrice}(x,y))$

This representation asserts that for all x, if x is an Apple Watch, then for all y, if y is a Samsung Watch, then x and y have the same price. It means that Apple and Samsung watches have the same price.

D. (5 points) From the sentence: “Cats are animals” we can deduce the goal “The tail of a cat is the tail of an animal”

a. Translate the above (the sentence and the goal) into first-order logic using the three predicates

- tail_of(x,y): x is the tail of y
- Cat(x): x is a cat

- Animal(x): x is an animal

Given Sentence: "Cats are animals"
This can be represented as: ∀x (Cat(x) → Animal(x))
This statement means that for all x, if x is a cat, then x is also an animal.

Goal: "The tail of a cat is the tail of an animal"
This can be represented as: ∀x (Cat(x) → (tail_of(x, cat) → tail_of(x, animal)))
This statement means that for all x, if x is a cat, then if x is the tail of a cat, it is also the tail of an animal.

So, in first-order logic:

∀x (Cat(x) → Animal(x)) represents the given sentence.
∀x (Cat(x) ^ tail_of(x, cat) → tail_of(x, animal))) represents the deduced goal.

B -Negate the goal and put all sentences in CNF.

Negate the goal.
Convert each sentence into CNF.
Combine all the CNF sentences using conjunctions (AND).
Let's start with the negation of the goal:

Original Goal (Negation): ¬(∀x (Cat(x) → (tail_of(x, cat) → tail_of(x, animal))))

Now, let's convert each sentence into CNF:

Given Sentence: ∀x (Cat(x) → Animal(x))
CNF Form: ¬Cat(x) ∨ Animal(x)

Negated Goal: ¬(∀x (Cat(x) → (tail_of(x, cat) → tail_of(x, animal))))
CNF Form: ∃x (Cat(x) ∧ ¬(tail_of(x, cat) → tail_of(x, animal)))

Now, we combine all the CNF sentences using conjunction (AND):

Final CNF Form: (¬Cat(x) ∨ Animal(x)) ∧ ∃x (Cat(x) ∧ ¬(tail_of(x, cat) → tail_of(x, animal)))

This is the CNF representation of the negated goal and the given sentence.